



SAMP RQ4.2 Concepts and formalisms landscape

What we have found for RQ4.2 so far for discussion. Nothing on this page is a final conclusion.



How to read this page

This page covers Problem 1 of RQ4.2, actor-side arrangements (see "Scope of this page" below).

The findings are kept at four levels:

- Research domain (D): a field that studies part of the problem, for example contract law.
- Concept (C): a distinction that the field gives us, for example unilateral termination.
- Representation requirement (MR): a capability the representation needs, condensed from several concepts, for example a governed lifecycle.
- Formalism (M): a concrete modeling language or method that could provide it, for example Symboleo.

Block 1: the research domains and the concepts they contributed.

Block 2: the eight representation requirements. The concepts behind them are on the subpages Block 2B (meaning and interpretation for SAMP) and Block 2A (detailed mapping with evidence).

Block 3: calibration of the search for modeling methods, in three passes. Its outcome is a search methodology for Block 4, not a candidate list. The evidence is on the Block 3 subpages.

References are given as [n]. The reference list is on the subpage  [SAMP RQ4.2. References](#).

Every statement is marked as a literature finding, our interpretation, or an open question.

The problem

A space program architecture (SPA) depends on arrangements between actors: a provider agrees to supply a launcher, an authority permits the use of a resource, a ministry approves a contract. These arrangements can change, and their change can alter the feasibility, the assumptions, or the configuration of the SPA.

Working term: Actor-mediated Architectural Dependence (AAD).

AAD = arrangement + changes to it + rules and conditions for change.

- **Arrangement:** what is currently established.
- **Change:** what can alter the arrangement.
- **Rules for change:** what determines when and how those changes can occur.

Scope of this page

RQ4.2 contains two complementary modeling problems.

- **Problem 1 actor-side arrangements:** how commitments, permissions, authority, membership, withdrawal, substitution, dependencies and lifecycle are represented and how they evolve.
- **Problem 2 collaborative design decisions:** how the participants in SPA design make, communicate, negotiate, revise and coordinate architecture decisions.

The distinction is between two modeling problems, not between two classes of actors. Problem 1 leads to concepts and methods for arrangements; Problem 2 to concepts and methods for collaborative decision-making. Some concepts and methods may contribute to both.

Everything on this page and its subpages analyzes Problem 1. The chain research domain → concept → characteristic → requirement → formalism was built for Problem 1 only.

Problem 2 has not yet been investigated. A low or absent score for a decision-oriented or agent-based method here says nothing about its value for Problem 2.

Synthesis question, to be answered after both problems are analyzed: can both be represented within one actor or organizational modeling approach, or are two coupled mechanisms required?

Scope: RQ4.2 Problem 1: actor-side arrangement modeling.

Characteristics of an arrangement (working list, provisional)

ID	Characteristic	Meaning
A1	External dependence	The arrangement is sustained or constrained by actors or institutions outside the program's unilateral control.
A2	Firmness	How binding the arrangement is, from expressed interest through agreement in principle to binding commitment.
A3	Interdependence	The arrangement depends on other arrangements and on the state of the SPA; a change in one propagates to others.
A4	Party dynamics / membership	Parties can enter, leave, merge, be replaced, or change what they provide.
A5	Lifecycle timing	Changes happen on the program timeline; the same change means something different before or after a milestone.
A6	Change origin	A change can come from a decision inside the program, from a decision by an outside actor, or from an event nobody decided.
A7	History dependence	What can still change depends on the current state or on what happened before; for example, after termination, cooperation may require a new arrangement.
A8	Change authority	Who may make each change: one party alone, several together, with outside approval, or subject to veto — and this can itself change.
A9	Automatic vs discretionary change	A condition either changes the arrangement by itself or only allows someone to decide.

Cross-cutting modeling dimensions (provisional)

The literature also exposed cross-cutting modeling dimensions that do not yet justify adding new AAD characteristics, but may need to be preserved in a future representation. They are shown separately from A1–A9.

ID	Dimension	Meaning
X1	State type	A represented state may be an intention, a program assumption, a social commitment, a permission, a capability, an observed fact, or an actual performance. These are not values of one generic arrangement state.
X2	Noncompliance and residual effects	A violation or breach can create consequences: repair, compensation, further obligations, obligations that survive termination. A relation changing or ending does not erase all consequences.
X3	Normative source, scope and precedence	A rule or institutional effect depends on which institution establishes it, where it applies, and which rule prevails in a conflict.
X4	Relation identity across change	After amendment, assignment, substitution, novation or replacement: does the same arrangement continue, or is a new one created?

Block 1. Research domains and their concepts

Question: which research fields have studied the actor-side phenomenon, and which distinctions do they give us?

What we did: two literature reviews, reading the sources in full where they were available.

What we found: 12 research domains, grouped in four families for convenience, and 109 concepts in total (listed one by one in Block 2B).

Result: the table below shows each domain, the concepts it contributed, and a provisional score on three questions. The domains are sources of concepts, not yet modeling choices.

Three provisional relevance dimensions (each measures a different thing; domains are not ranked)

- Problem framing: how strongly the domain explains the structural problem SAMP faces.
- Semantic contribution: how strongly the domain supplies distinctions a correct representation of AAD may need to preserve (A1–A9 and X1–X4).
- Modeling bridge: how strongly the domain offers a plausible path to a formal or engineering representation that could connect to the SPA.

Scores: ●●● High · ●● Medium · ● Limited. These are provisional and are not rankings of candidate solutions.

Research domain	Concepts found	Problem framing	Semantic contribution	Modeling bridge	Note
NORMATIVE AND INSTITUTIONAL	binding force; obligations; rules and who may change them				
D1 Contract and legal relations (private contracts, treaties, financial contracts)	Being bound: intention to be bound; binding vs non-binding; preliminary agreement; hard	●●	●●●	●●	Legal literature. The most detailed vocabulary for binding, ending and authority.

Research domain	Concepts found	Problem framing	Semantic contribution	Modeling bridge	Note
	vs soft law; precision; consent. Changing and ending: unilateral termination; suspension vs termination; amendment; changed circumstances. Authority and positions: obligation; delegation; proposal / veto rights; legal power to change a legal relation; automatic vs optional legal effect. Identity and transfer: agreement vs document vs contract; assignment; substitution; novation. [1][2][3][4][5][6][7][8][34]				The bridge is through computational contract languages, not legal doctrine itself.
D2 Social commitments	Directed commitment (debtor → creditor); commitment content; conditional commitment; commitment lifecycle (create, detach, discharge, cancel, release, replace, violate, expire); deadline; achievement vs maintenance commitment. [23][34]	••	•••	•••	A directed relation between two parties; distinct from dependence, and not in itself a legal contract. Formal commitment lifecycles exist.
D3 Institutional theory,	Rules: rules-in-use; regulative vs	•••	•••	••	Rules and how rules change.

Research domain	Concepts found	Problem framing	Semantic contribution	Modeling bridge	Note
institutional analysis	constitutive rules; operational, collective-choice and constitutional rules; rules of change; action situation; boundary / position / choice / control rules; rule statement (attributes, deontic, aim, condition, or-else). Change of rules: drift, displacement, layering, conversion; rules about rules. Time: path dependence; lock-in; timing and sequence. [15][16][17][18][34]				Partly formalized (the Institutional Grammar).
D4 Organizational / normative MAS and social action	Autonomous agent; social dependence; goal delegation and adoption; role; role enactment; social contract; institutional act; constitutive rules; institutional power; governed state change. [23][24][34]	••	•••	•••	Agents acting within institutions. The computational treatment of institutional power sits here.
D5 Deontic logic, normative reasoning	Obligation; permission; prohibition; conditional obligation; strong vs weak permission; defeasibility; contrary-to-duty	•	••	•••	The reviewed sources cover what is required or allowed, not the institutional power to change the

Research domain	Concepts found	Problem framing	Semantic contribution	Modeling bridge	Note
	(violation-triggered) obligation; derogation. [19][20][21][22]				norms. Formal by nature.
RELATIONAL AND ORGANIZATIONAL	dependence between organizations; who takes part				
D6 Collaborative networks, virtual organizations	Temporary collaboration; partner membership and its ambiguity; partner competence and capacity; task allocation; partner addition / removal; reconfiguration; terms of participation. [14][34]	••	••	••	The configuration and reconfiguration of a collaboration. Covers membership, which the normative domains barely touch.
D7 Organization studies, inter-organizational relations	Dependence: resource dependence; external control; structural interdependence. Membership: alliance lifecycle and dissolution; partner entry / exit; rewiring of ties; open membership. Change: disruption by external shocks; contractual vs relational governance; ecosystem alignment. [9][10][11][12][13]	•••	••	•	Firms and alliances as parties. Mostly descriptive and empirical; no formal model of actor arrangements found in the reviewed work.
D8 Goal-oriented and sociotechnical requirements	Dependency: depender, dependee, dependum; goal / task / resource	••	••	•••	Dependence and assumption, which do not imply a

Research domain	Concepts found	Problem framing	Semantic contribution	Modeling bridge	Note
	dependency. Intentional state: goal; assumption and domain assumption; distinction from social commitment. Constraints: temporal and data constraints on goals and tasks. [34]				commitment. Modeling languages exist for these dependencies.
TIMING AND COMMITMENT	when to commit; how change is controlled				
D9 Program and project management, engineering change control	Lifecycle: phases (0, A-F); milestones; reviews; stage / gate; go / kill / hold / recycle; must-meet vs should-meet criteria; staged commitment. Change control: configuration baseline; change request; change class; configuration control board; classified approval; acceptance review. [27][28][29][30]	•••	••	••	Practice and standards; native to the space domain. In the ECSS change-control process examined, every change requires an approval decision; no automatic transition was found.
D10 Real options, investment under uncertainty	Irreversible investment; sunk cost; option value of waiting; option to defer; option to abandon; staged investment; delayed commitment (the source's example: postponing a	••	••	•	When to commit. Says nothing about how binding an agreement is. Valuation methods, not representation.

Research domain	Concepts found	Problem framing	Semantic contribution	Modeling bridge	Note
	design freeze); right but not obligation. [25] [26]				
ENGINEERING CONTEXT	the systems side				
D11 Systems of systems	Operational and managerial independence of constituents; evolutionary development; reconfiguration; distributed ownership and control; SoS types by degree of central authority (directed, acknowledged, collaborative, virtual); constituents joining and leaving; agreement processes in SoS standards. [31] [32][34]	•••	•	••	Explains the setting that makes actor-side dependence matter and is the likely engineering bridge to the SPA. The reviewed SoS literature does not itself provide semantics for agreements, commitments or change authority.
D12 Other supporting concept sources	Fault taxonomy: cause (natural vs human-made) × intent (deliberate vs not); action theory: agent vs event causation. [33][34]	•	•	•	Used only to define change origin.

Research domains are grouped for convenience. References are in [].



Finding across domains. Several domains distinguish kinds of state: what an actor intends, what the program assumes, what one party owes another, what is institutionally permitted, and what is actually true or delivered. These are not one "arrangement state". This is dimension X1.

Block 2. Provisional representation requirements

Question: what would a representation of actor-side arrangements need to be able to express?

What we did: the 109 concepts from Block 1 are too many to compare modeling methods against one by one. We condensed them into eight representation requirements (MR1–MR8). Each requirement is a synthesis of several concepts and points back to them.

Result: the table below. The concept-level material stays on the subpages: Block 2B gives each concept's meaning and our interpretation for SAMP; Block 2A gives the detailed mapping with evidence.



Read this as a visualization, not as the level of analysis.

Provisional representation requirements are a high-level synthesis for visualization and discussion. They combine findings across multiple concept families and were not evaluated directly as independent units. The underlying analysis remains at concept level.

- Concept interpretation (what each concept means for SAMP) at:  [SAMP RQ4.2. Block 2B: Concept Interpretation for SAMP](#)
- Detailed evidence mapping (● / ○ coverage, evidence levels, references) at:  [SAMP RQ4.2. Block 2A: Detailed Concept × AAD Mapping](#)

To challenge a requirement: follow its concept IDs to Block 2B, then to Block 2A.

ID	Provisional representation requirement	Justification	Derived from
MR1	An arrangement as a first-class element. Parties with their capacities, a subject, and terms; distinct from the technical relations it sustains and from any document that records it.	Without it, an arrangement can only appear as an attribute of a technical element, and its parties and terms are lost.	C1.17 · C1.18 · C2.1 · C2.2 · C4.5 · C6.7 · C8.2 · A1 External dependence · A2 Firmness · X1 State type · X4 Relation identity across change
MR2	Typed and graded state. Intention, program assumption, commitment, permission or obligation, and fact are different state types; within the normative type, how binding an arrangement is can be graded.	The technology-availability case: "provider plans T", "program assumes T", "provider committed to T" and "T delivered" are different states.	C1.1 · C1.2 · C1.3 · C2.7 · C4.3 · C5.1 · C8.3 · C8.4 · C8.5 · C10.8 · A2 Firmness · X1 State type
MR3	Governed lifecycle on the program timeline. Named transitions (establish, revise, suspend, resume, terminate, replace, violate, expire); which transitions are admissible depends on the current state and on history; violation and obligations that survive termination are states; the effect of a transition depends on when it occurs relative to milestones.	An available / unavailable flag cannot tell suspension from termination, or a change before a milestone from the same change after it.	C1.6 · C1.7 · C1.8 · C1.15 · C1.16 · C2.4 · C2.5 · C2.6 · C3.9 · C3.10 · C4.8 · C5.5 · C5.8 · C7.4 · C9.1 · C9.2 · C9.8 · C9.12 · C10.1 · C10.2–C10.6 · A5 Lifecycle timing · A7 History dependence · X2 Noncompliance and residual effects · X4 Relation identity across change
MR4	Authority and power per transition. For each transition: who may initiate, who must agree, who may block or approve; permission to act	Who can change an arrangement decides how exposed the SPA is to that change.	C1.5 · C1.10 · C1.11 · C1.12 · C1.14 · C3.1 · C3.3 · C3.4 · C3.6 · C4.6 · C4.7 · C5.7 · C7.2 · C9.7 · C9.10 · C9.11 ·

ID	Provisional representation requirement	Justification	Derived from
	distinct from the power to make the change effective; authority itself changeable through rules at a higher level.		A8 Change authority · X3 Normative source, scope and precedence
MR5	Triggered versus discretionary change. A condition either causes a transition by itself or opens a decision for an entitled actor; both forms occur (knock-out criteria, activation conditions, automatic legal effects, options).	An automatic effect can be predicted from the condition; a discretionary one depends on someone's decision.	C1.9 · C1.13 · C2.3 · C3.7 · C5.2 · C5.3 · C9.4 · C9.5 · C10.7 · A9 Automatic vs discretionary change · A6 Change origin
MR6	Dependencies among arrangements and with the SPA. An arrangement can be conditional on other arrangements, on norms, and on the state of the architecture; a change propagates along these dependencies in both directions.	This is where the actor side meets the technical architecture (working interpretation of RQ4.3, below).	C1.14 · C1.21 · C5.5 · C5.6 · C6.3 · C7.3 · C7.6 · C8.1 · C8.6 · C9.9 · C11.4 · A3 Interdependence
MR7	Dynamic parties. Actors enter, leave, merge, are substituted, or change what they can provide; role, performer, party and liability can change separately; whether the arrangement keeps its identity under substitution is explicit.	Replacing a provider is not one operation; the program must know what moves and what stays.	C1.19 · C1.20 · C3.6 · C4.4 · C6.2–C6.6 · C7.5–C7.7 · C11.2 · C11.7 · A4 Party dynamics / membership · X4 Relation identity across change
MR8	Origin and normative source of change. Each change event is attributed to a party's decision, an outside actor's decision, a non-agentive event, or a shift in the governing rules; the source of a rule and the precedence between rules are recorded.	A sanction, a bankruptcy and a program decision call for different responses even when their effect is the same.	C1.9 · C3.2 · C3.8 · C3.11 · C5.4 · C7.8 · C11.6 · C12.1 · C12.2 · A6 Change origin · X3 Normative source, scope and precedence

No requirement covers an AAD characteristic completely. No formalism is expected to satisfy a requirement completely. A concept can feed several requirements, and a requirement draws on several concepts.

Working interpretation of RQ4.3 (open question)

RQ4.3 asks how the actor-level model can be coupled to the technical architecture.

My current understanding of this question:

- A change on the actor side can change the architecture. Example: a provider withdraws, an authorization is revoked, a commitment is reduced. The architecture may lose an assumption, a constraint, a feasible option.
- A choice in the architecture can create a need on the actor side. Example: choosing a launcher creates a need for a supply contract, a license, or a permission.
- The model should be able to answer: why is this architecture feasible right now, and which actor-side conditions does that depend on?

Whether the coupling should work in both directions is not decided.

In Block 3 we keep two questions apart: how well a formalism represents the actor side (RQ4.2), and how well it supports this reading of RQ4.3.

Block 3. Calibrating the search for modeling methods

We used several search-and-screening passes to work out how the actual literature search should be done. Each pass exposed a weakness in the one before. Block 3 was therefore not meant to produce the final candidate set. It was used to tune the search space, the vocabulary and the screening procedure before the real search.

The basic procedure, repeated in each pass

- **3A. Define what we search for:** derive search directions from the concepts and representation needs of Blocks 1 and 2.
- **3B. Search:** use those directions to find modeling methods, languages, formalisms, frameworks and the research communities behind them.
- **3C. Screen and inspect:** check what was found, what was missed, and whether the search framing itself has to change.

Pass 1: initial search framing

Question: does the first search map give broad coverage?

What we did: used 14 search topics, grouped by judgment from the requirements; ran a first scoping search, mostly general web search; screened what came back.

What we learned:

- the search depended heavily on the wording of the 14 topics;
- some methods were set aside too quickly as "represented" by another method;
- whole research communities were probably being missed.

Historical calibration numbers, not results: 133 items found, 115 after merging versions, 41 kept.

Details →  [SAMP RQ4.2. Block 3A: Search Topics](#) ·  [SAMP RQ4.2. Block 3B, Pass 1: Scoping Search for Candidates](#) ·  [SAMP RQ4.2. Block 3C, Pass 1: Candidate Screening and Classification](#)

Pass 2: false-negative challenge

Question: what might Pass 1 have missed or screened out too early?

What we changed: searched from single concept distinctions instead of topics; used other communities' terminology; looked into neighboring fields; re-checked every item Pass 1 had set aside as "represented by" another method.

What we learned:

- relevant research lines appeared that Pass 1 had not reached;
- the choice of search words decides which community you reach;
- one pass of discovery and screening is not enough.

Details → [SAMP RQ4.2. Block 3B, Pass 2: Searching for What Pass 1 Missed](#) · [SAMP RQ4.2. Block 3C, Pass 2: Screening What Pass 1 Missed](#)

Pass 3: auditing and calibrating the search space

Question: was the search space itself defined correctly?

What we changed:

- two independent blind sessions regrouped all 109 concepts from scratch, without seeing the 14 topics;
- we compared the two blind groupings and combined them;
- we compared the combined result with the old 14 topics, concept by concept;
- we searched the gaps and different modeling traditions, and recorded every query;
- we tested, with a check fixed in advance, whether the revised terms reach research neighborhoods the earlier passes missed.

What we learned:

- several useful search directions were missing from the original framing, for example staged reviews, configuration control, real options, decision rights and relation identity;
- some old topics were too broad or mixed different mechanisms;
- new communities appeared: ontology engineering, engineering change and DSM, engineering systems and design flexibility, computational trust, runtime norm monitoring, value modeling, and holistic system-of-systems architecting;
- different fields use very different words for the same modeling problem;
- generic search terms now reach the stakeholder, architecture design-space and holistic system-of-systems neighborhood. That neighborhood belongs mainly to RQ4.3 and Problem 2.

Details → [SAMP RQ4.2. Block 3, Pass 3, Step B: Comparison of the Two Blind Groupings](#) · [SAMP RQ4.2. Block 3, Pass 3, Steps B1–B2: Combined Blind Map and Audit of the 14 Search Topics](#) · [SAMP RQ4.2. Block 3, Pass 3, Step C: Search Map and Search Plan](#) · [SAMP RQ4.2. Block 3, Pass 3, Step C: Calibration Search Record and Block-4 Search Protocol](#)



Block 3 outcome

The outcome of Block 3 is a calibrated search methodology, not a candidate shortlist.

All Block-3 findings are preserved on the subpages.

Block-3 candidate IDs and statuses are historical calibration records only.

Block 4 starts from zero.

A method enters the candidate set only if the Block-4 search finds it again.

Block 3 informs the vocabulary and the search design. It does not populate Block 4.

Search methodology for Block 4

High-level only. The search has not started.

1. Search directions. Use the concept-derived search directions calibrated in Pass 3.

2. Two ways into each direction.

- by the phenomenon or modeling capability we need;
- by the terms the relevant research community itself uses.

3. Several modeling lenses. Search not only by capability, but also by:

- kind of model: organizational, institutional, stakeholder, actor and dependency, enterprise architecture, governance, policy, value network;
- representation mechanism: ontology or metamodel, logic or rules, state or transition model, process, workflow or Petri net, graph or network model.

4. Several scholarly sources. General web search alone is not enough.

- Scopus and/or Web of Science where available;
- Google Scholar;
- DBLP for computer science;
- law sources such as HeinOnline or SSRN;
- web search only as a supplement, for tools, standards and project pages.

5. Record every search. Per query: search direction, exact query, source, date, results examined, artifacts found, research community and terminology, reason for inclusion or exclusion.

6. Separate registers. The Problem 1 candidate search is the main search. Results that belong mainly to Problem 2, or to RQ4.3 and the coupling to the architecture, go into their own registers instead of being discarded.

7. Snowballing. Backward and forward snowballing starts from artifacts and reviews found again in Block 4, not from the Block-3 candidate list.

8. Independence from Block 3. No Block-3 candidate ID or status is carried forward. A method found in Block 3 that appears again in Block 4 gets a new Block-4 ID and Block-4 provenance.

The detailed protocol drafted at the end of Pass 3 →  [SAMP RQ4.2. Block 3, Pass 3, Step C: Calibration Search Record and Block-4 Search Protocol](#)

RQ4.2 Problem 2. Methods about making, negotiating or coordinating decisions that surfaced during Block 3 are kept on a separate list for the later Problem 2 work. The list →  [SAMP RQ4.2. Block 3C, Pass 1: Candidate Screening and Classification](#)

Block 4. Systematic search for modeling methods



Block 4A: protocol frozen (revision 3, approved 29 September 2026). Next: Block 4B search execution, by a deterministic pipeline under construction. No Block-4 searches have been run.

Frozen protocol and query bank →  [SAMP RQ4.2. Block 4A: Frozen Systematic Search Protocol, revision 3 \(APPROVED, frozen 29 Sep 2026\).](#)

Provenance note

An earlier exploratory Block 3 pass is retained separately for provenance. Its candidate list, numbering and synthesis were produced before systematic candidate discovery and are not used in the canonical analysis. →  [SAMP RQ4.2. Legacy exploratory Block 3](#)

-  [SAMP RQ4.2. References](#)
-  [SAMP RQ4.2. Block 2A: Detailed Concept × AAD Mapping](#)
-  [SAMP RQ4.2. Block 2B: Concept Interpretation for SAMP](#)
-  [SAMP RQ4.2. Block 3A: Search Topics](#)
-  [SAMP RQ4.2. Block 3B, Pass 1: Scoping Search for Candidates](#)
-  [SAMP RQ4.2. Block 3C, Pass 1: Candidate Screening and Classification](#)
-  [SAMP RQ4.2. Block 3D: Detailed Evaluation of Shortlisted Candidates](#)
-  [SAMP RQ4.2. Legacy exploratory Block 3](#)
-  [SAMP RQ4.2. Block 3B, Pass 2: Searching for What Pass 1 Missed](#)
-  [SAMP RQ4.2. Block 3C, Pass 2: Screening What Pass 1 Missed](#)
-  [SAMP RQ4.2. Block 3, Pass 3, Step B: Comparison of the Two Blind Groupings](#)
-  [SAMP RQ4.2. Block 3, Pass 3, Steps B1–B2: Combined Blind Map and Audit of the 14 Search Topics](#)
-  [SAMP RQ4.2. Block 3, Pass 3, Step C: Search Map and Search Plan](#)
-  [SAMP RQ4.2. Block 3, Pass 3, Step C: Calibration Search Record and Block-4 Search Protocol](#)
-  [SAMP RQ4.2. Block 4A: Frozen Systematic Search Protocol \(draft for review\)](#)
-  [SAMP RQ4.2. Block 4A: Frozen Systematic Search Protocol, revision 3 \(APPROVED, frozen 29 Sep 2026\)](#)